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## Automating Incident Response with the Power of Kusto

Bert-Jan Pals

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Learn | Share | Practice

# Bert-Jan Pals



Defensive Security Expert



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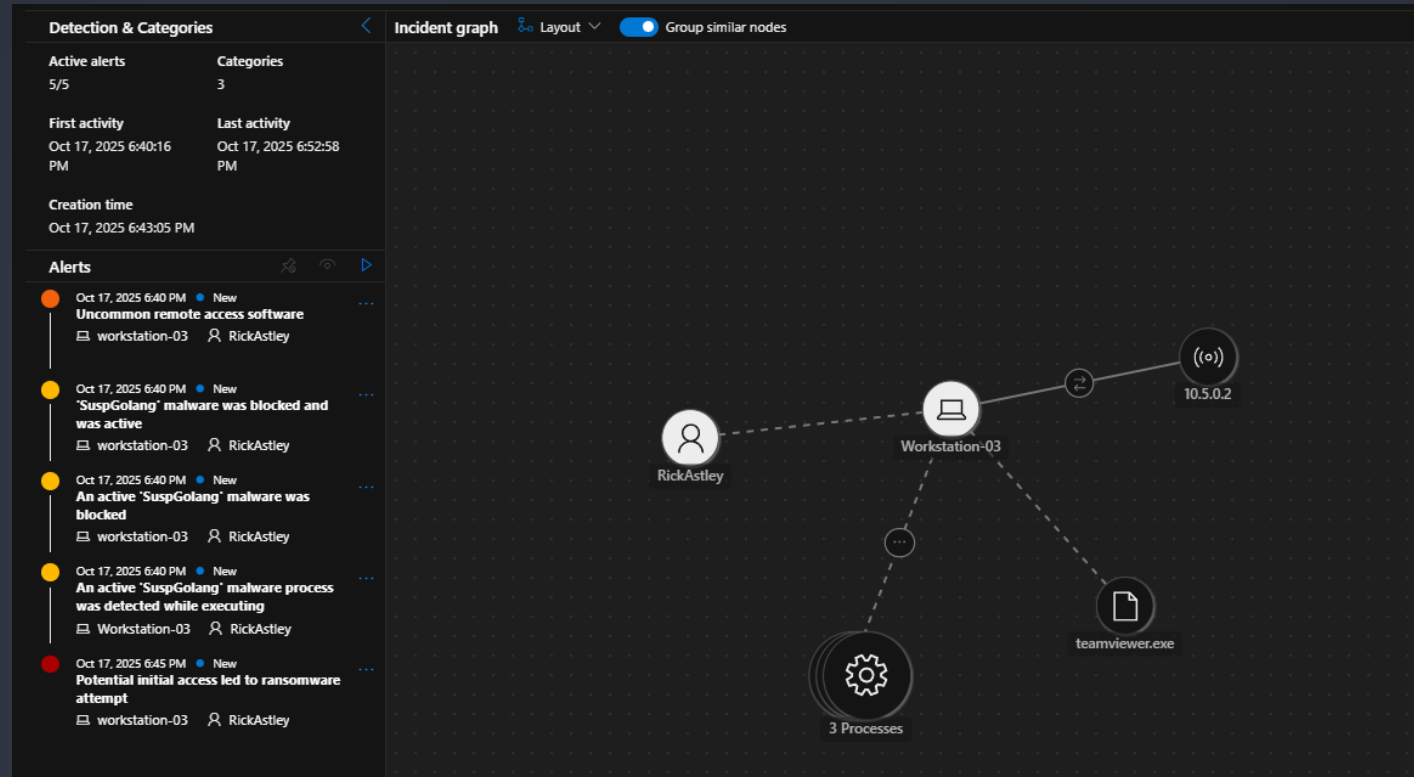
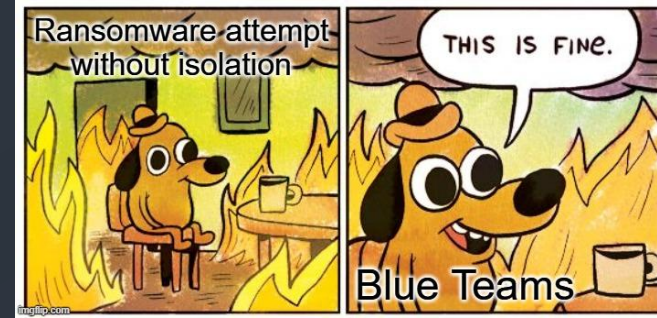


<https://kqlquery.com/>

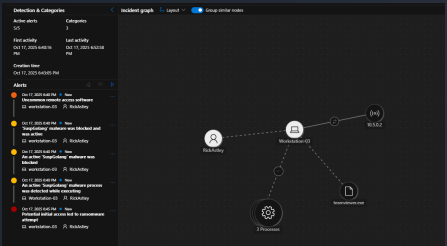
# Agenda

- Why Automate?
- Response Actions
- Custom Detection Rules
- Workbooks
- Logic Apps
- APIs
- KustoHawk 🦅

# Why do you want to automate?



Initial Alert

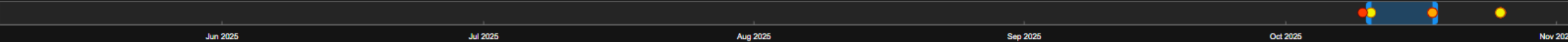


Workstation-03

Copilot View in map Device value Set criticality ...

High Criticality: Low Active ExcludedAttackDisruption ExclusionGroup - AttackDisruption

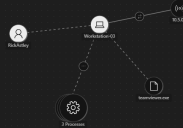
Overview Incidents and alerts Timeline Configuration management Security recommendations Inventories Discovered vulnerabilities Missing KBs



Highlighted alert: An active 'SuspGolang' malware was blocked

Export Search Oct 10, 2025 6:52:58 PM - Oct 17, 2025 6:52:58 PM Customize columns Timeline settings Filter

| Event time                  | Event                                                                       | Additional information   | Service source                  | User                | Entities                          |
|-----------------------------|-----------------------------------------------------------------------------|--------------------------|---------------------------------|---------------------|-----------------------------------|
| Load newer results          |                                                                             |                          |                                 |                     |                                   |
| Oct 17, 2025 6:52:58.860 PM | An active 'SuspGolang' malware was blocked                                  | Malware                  |                                 |                     |                                   |
| Oct 17, 2025 6:52:58.860 PM | 'SuspGolang' malware was blocked and was active                             | Malware                  |                                 |                     |                                   |
| Oct 17, 2025 6:52:58.860 PM | Defender detected and quarantined 'Trojan:Win32/SuspGolang.AG' in file '... | Malware                  | Microsoft Defender for Endpoint | AzureAD\RickAstley  | explorer.exe > teamviewer.exe     |
| Oct 17, 2025 6:52:58.802 PM | setup.exe created file dxcompiler.dll                                       |                          | Microsoft Defender for Endpoint | nt authority\system | MicrosoftEdge_X64_141.0.3511.1000 |
| Oct 17, 2025 6:52:58.699 PM | setup.exe created file dual_engine_adapter_x64.dll                          |                          | Microsoft Defender for Endpoint | nt authority\system | MicrosoftEdge_X64_141.0.3511.1000 |
| Oct 17, 2025 6:52:58.667 PM | setup.exe created file d3dcompiler_47.dll                                   |                          | Microsoft Defender for Endpoint | nt authority\system | MicrosoftEdge_X64_141.0.3511.1000 |
| Oct 17, 2025 6:52:58.632 PM | setup.exe deleted cookie_exporter.exe                                       | T1070.004: File Deletion | Microsoft Defender for Endpoint | NT AUTHORITY\system | MicrosoftEdgeUpdate.exe > ...     |
| Oct 17, 2025 6:52:58.631 PM | setup.exe created file cookie_exporter.exe                                  |                          | Microsoft Defender for Endpoint | nt authority\system | MicrosoftEdge_X64_141.0.3511.1000 |

[illegible]

The screenshot shows the 'Workstation-03' interface. At the top, there's a navigation bar with 'Hosts', 'Incidents and alerts', 'Timeline', 'Configuration management', 'Security recommendations', 'Hosts tree', 'Discovered vulnerabilities', and 'Missing OS'. Below this is a timeline view showing events from July 2023 to May 2024. A specific event is highlighted: 'Active Directory system was blocked' on July 10, 2023, at 15:02:55. The event details show a warning icon, a description of the system being blocked, and a list of related events. The 'Related events' section lists several events related to the blocked system, including 'Microsoft Security to inspect' and 'Microsoft Security to inspect'.

New query\* X | New query\* X | New query\* X | New query X | tse\* X | New query\* X | New query\* X | New query\* X | New query\* X | New query\* X | New query\* X | New query\* X | ...

Getting startedResultsQuery history

Export Show empty columns

14 items Search

00:01.263 Low

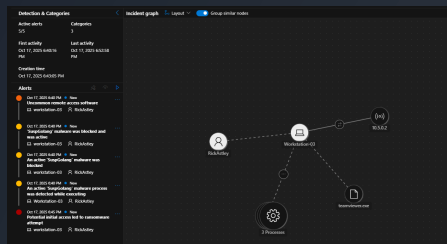
Chart type

Full screen

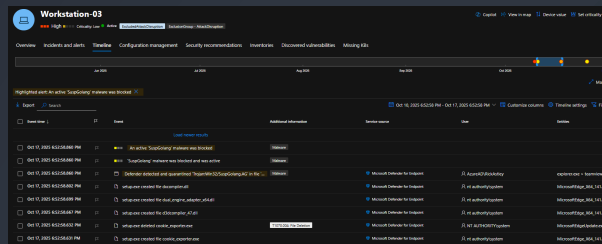
Filters: Add filter

|                          | TimeGenerated | Stable                | Timestamp    | ActionType              | DeviceName                | Type           | AccountName  | AccountDomain | ReportId_Jong | AdditionalFields             | DeviceId                      |                       |
|--------------------------|---------------|-----------------------|--------------|-------------------------|---------------------------|----------------|--------------|---------------|---------------|------------------------------|-------------------------------|-----------------------|
| <input type="checkbox"/> | >             | Oct 17, 2025 6:40:... | DeviceEvents | Oct 17, 2025 6:40:18 PM | OtherAlertRelatedActivity | Workstation-03 | DeviceEvents |               | 2382          | {"Description": "Defender... | 2694a7cc2225f3b66f...         |                       |
| <input type="checkbox"/> | >             | Oct 17, 2025 6:40:... | DeviceEvents | Oct 17, 2025 6:40:18 PM | OtherAlertRelatedActivity | Workstation-03 | DeviceEvents |               | 2381          | {"Description": "Defender... | 2694a7cc2225f3b66f...         |                       |
| <input type="checkbox"/> | >             | Oct 25, 2025 2:21:... | DeviceEvents | Oct 25, 2025 2:21:39 PM | OtherAlertRelatedActivity | Workstation-03 | DeviceEvents | RickAstley    | AzureAD       | 45095                        | {"Description": "WmiPrivS...  | 2694a7cc2225f3b66f... |
| <input type="checkbox"/> | >             | Oct 25, 2025 2:18:... | DeviceEvents | Oct 25, 2025 2:18:06 PM | OtherAlertRelatedActivity | Workstation-03 | DeviceEvents | RickAstley    | AzureAD       | 44967                        | {"Description": "schtasks.... | 2694a7cc2225f3b66f... |
| <input type="checkbox"/> | >             | Oct 25, 2025 2:18:... | DeviceEvents | Oct 25, 2025 2:18:06 PM | OtherAlertRelatedActivity | Workstation-03 | DeviceEvents | RickAstley    | AzureAD       | 44967                        | {"Description": "schtasks.... | 2694a7cc2225f3b66f... |
| <input type="checkbox"/> | >             | Oct 25, 2025 1:33:... | DeviceEvents | Oct 25, 2025 1:33:32 PM | OtherAlertRelatedActivity | Workstation-03 | DeviceEvents | RickAstley    | AzureAD       | 39696                        | {"Description": "schtasks.... | 2694a7cc2225f3b66f... |

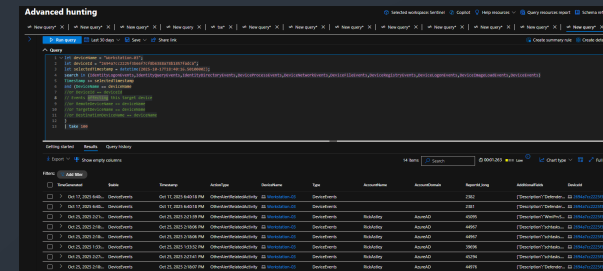
## Initial Alert



## Timeline investigation



## Run hunting queries



Advanced hunting > Workstation-03



## Workstation-03

High Criticality: Low Active ExcludedAttackDisruption ExclusionGroup - AttackDisruption

Copilot View in map Device value

Overview Incidents and alerts Timeline Configuration management Security recommendations Inventories Discovered vulnerabilities Missing KBs

### VM details

| Category             | Type                                              |
|----------------------|---------------------------------------------------|
| Computers and Mobile | Unknown                                           |
| Subtype              | Discovery sources                                 |
| Workstation          |                                                   |
| Domain               | OS                                                |
| AAD joined           | Windows 11 64-bit (Release 24H2 Build 26100.4652) |
| SAM name             | Health state                                      |
| -                    | Active                                            |
| Data sensitivity     | IP addresses                                      |
| None                 | Private: 192.168.178.144<br>Public: 84.26.17.125  |

### Active alerts (Last 100 days)

## Risk

91 active

Active alert

High

View alert

Device

Full scan status is unknown

Type State Date & time

## Isolate Computer Workstation-03

This action will isolate the device from the network. It will remain connected to the Microsoft Defender for Endpoint service.

☐ Use isolation exclusion to allow specific communication while the device is isolated.

Comment:

① You can undo device isolation through the Actions menu.

Confirm

Close

### Logged on users (Last 30 days)

## 1 logged on user

### Most logons

rick.astley  
Local admin

### Least logons

rick.astley  
Local admin

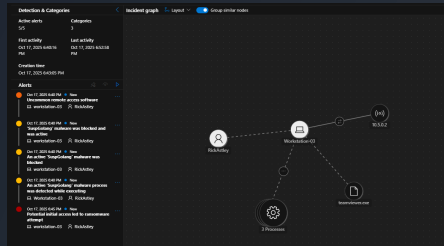
### Newest logon

rick.astley  
Local admin

View 1 logged on user



## Initial Alert



## Timeline investigation

The screenshot shows a timeline investigation in a security dashboard. The table lists events with columns for 'Time', 'Event', 'Additional information', 'Status', and 'Details'. The events are filtered by 'Workstation-03' and 'R102'.

| Time                   | Event                    | Additional information   | Status  | Details                  |
|------------------------|--------------------------|--------------------------|---------|--------------------------|
| 04/10/2023 10:00:00 PM | Workstation-03 logged on | Workstation-03 logged on | Success | Workstation-03 logged on |
| 04/10/2023 10:00:00 PM | Workstation-03 logged on | Workstation-03 logged on | Success | Workstation-03 logged on |
| 04/10/2023 10:00:00 PM | Workstation-03 logged on | Workstation-03 logged on | Success | Workstation-03 logged on |
| 04/10/2023 10:00:00 PM | Workstation-03 logged on | Workstation-03 logged on | Success | Workstation-03 logged on |
| 04/10/2023 10:00:00 PM | Workstation-03 logged on | Workstation-03 logged on | Success | Workstation-03 logged on |

## Run hunting queries

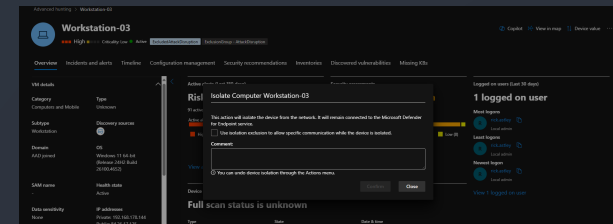
The screenshot shows a hunting query in a security dashboard. The table lists results with columns for 'Time', 'Event', 'Additional information', 'Status', and 'Details'. The results are filtered by 'Workstation-03' and 'R102'.

| Time                   | Event                    | Additional information   | Status  | Details                  |
|------------------------|--------------------------|--------------------------|---------|--------------------------|
| 04/10/2023 10:00:00 PM | Workstation-03 logged on | Workstation-03 logged on | Success | Workstation-03 logged on |
| 04/10/2023 10:00:00 PM | Workstation-03 logged on | Workstation-03 logged on | Success | Workstation-03 logged on |
| 04/10/2023 10:00:00 PM | Workstation-03 logged on | Workstation-03 logged on | Success | Workstation-03 logged on |
| 04/10/2023 10:00:00 PM | Workstation-03 logged on | Workstation-03 logged on | Success | Workstation-03 logged on |
| 04/10/2023 10:00:00 PM | Workstation-03 logged on | Workstation-03 logged on | Success | Workstation-03 logged on |

## Challenges

- Time to contain
- Standardized approach for hunting queries/timeline investigation
- There is so much data available, where should one begin to look?
- Everything looks suspicious if you start looking
- Incident efficiency -> manual actions

## Containment



# Response Actions

| Devices                       | Files           | Users                    | Emails          |
|-------------------------------|-----------------|--------------------------|-----------------|
| Isolate                       | Allow/Block     | Mark user as compromised | Move to mailbox |
| Collect Investigation Package | Quarantine file | Disable user             | Delete email    |
| Run antivirus scan            |                 | Force password reset     |                 |
| Initiate investigation        |                 |                          |                 |
| Restrict app execution        |                 |                          |                 |

# Manual Response

 **simulations**

High

Criticality: Low

Internet facing

Workstations

Overview

Incidents and alerts

Timeline

Configuration management

Security recommendations

Inventories

Discovered vulnerabilities

Missing KBs

VM details

Category

Computers and Mobile

Type

Unknown

Subtype

Workstation

Discovery sources



Domain

Workgroup

OS

Windows 11 64-bit  
(Release 24H2 Build 26100.6584)

SAM name

-

Health state

Inactive

Data sensitivity

None

IP addresses

10.0.0.4  
[See IP addresses info](#)

MAC address

7C-ED-8D-4B-45-6E

First seen

Oct 5, 2025 2:45:25 PM

Active alerts (Last 180 days)

**Risk level: High**

43 active alerts, 1 active incident

Active alerts

High (6)

Medium (23)

Low (9)

1 more

[View all incidents and alerts](#)

Security assessments

**Exposure level: Medium**

68 active security recommendations

Discovered vulnerabilities (126)

Critical (1)

High (85)

2 more

[View all recommendations](#)

Logged on users (Last 30 days)

**0 logged on users**

There were no logged on users during the given time range

Device health status

**Security intelligence update status is unknown +3 more issues**

Copilot

View in map

Device value

Set criticality

...

Manage tags

Report device inaccuracy

Run Antivirus Scan

Collect Investigation Package

Restrict App Execution

Initiate Automated Investigation

Initiate Live Response Session

Isolate Device

Ask Defender Experts

Collect Support Logs

Action center

Download force release from isolation script

Exclude

Go hunt

Turn on troubleshooting mode

Policy sync

# Device Actions

Required fields: DeviceId

```
1 let ImageLoads = DeviceImageLoadEvents
2 | where ActionType == 'ImageLoaded'
3 | where FileName =~ "samlib.dll"
4 | where isnotempty(InitiatingProcessSHA256)
5 | invoke FileProfile(InitiatingProcessSHA256, 1000)
6 | where GlobalPrevalence <= 50 or isempty(GlobalPrevalence)
7 | project Timestamp, DeviceId, DeviceName, ActionType, FileName
8 let NamedPipes = DeviceEvents
9 | where ActionType == 'NamedPipeEvent'
10 | where isnotempty(InitiatingProcessSHA256)
11 | join kind=inner (ImageLoads | distinct InitiatingProcessSHA256)
12 | where parse_json(AdditionalFields).PipeName == @"\\Device\\Name"
13 | project Timestamp, DeviceId, DeviceName, ActionType, FileName
14 let Connection = DeviceNetworkEvents
15 | where ActionType == "ConnectionSuccess"
16 | where isnotempty(InitiatingProcessSHA256)
17 | join kind=inner (ImageLoads | distinct InitiatingProcessSHA256)
18 | project Timestamp, DeviceId, DeviceName, ActionType, RemoteIP
19 union NamedPipes, ImageLoads, Connection
20 | sort by Timestamp asc, DeviceId, InitiatingProcessSHA256
21 | scan with_match_id=Id declare (Step:string, Delta:timespan) w
22   step InitialConnection: ActionType == "ConnectionSuccess" =
23   step NamedPipe: ActionType == 'NamedPipeEvent' and DeviceId
24   step ImageLoad: ActionType == 'ImageLoaded' and DeviceId ==
25 )
26 | where Step == 's3'
27
```

| Getting started                     | Results                 | Query                 | Take actions       |
|-------------------------------------|-------------------------|-----------------------|--------------------|
| Export                              | Link to incident        | Take actions          | Show empty columns |
| Filters:                            | Add filter              |                       |                    |
| <input checked="" type="checkbox"/> | Timestamp               | DeviceId              | DeviceName         |
| <input checked="" type="checkbox"/> | > Oct 6, 2025 5:57:5... | 2694a7cc2225f3b66f... | workstation-03     |
| <input checked="" type="checkbox"/> | > Oct 9, 2025 6:06:2... | 2694a7cc2225f3b66f... | workstation-03     |
| <input checked="" type="checkbox"/> | > Oct 17, 2025 6:41:... | 2694a7cc2225f3b66f... | workstation-03     |
| <input checked="" type="checkbox"/> | > Oct 25, 2025 2:18:... | 2694a7cc2225f3b66f... | workstation-03     |

## Advanced hunting

New query\* X | New query\* X |

Run query Set in query

### Query

```
1 let ImageLoads = DeviceImageLoadEvents
2 | where ActionType == 'ImageLoaded'
3 | where FileName =~ "samlib.dll"
4 | where isnotempty(InitiatingProcessSHA256)
5 | invoke FileProfile(InitiatingProcessSHA256, 1000)
6 | where GlobalPrevalence <= 50 or isempty(GlobalPrevalence)
7 | project Timestamp, DeviceId, DeviceName, ActionType, FileName
8 let NamedPipes = DeviceEvents
9 | where ActionType == 'NamedPipeEvent'
10 | where isnotempty(InitiatingProcessSHA256)
11 | join kind=inner (ImageLoads | distinct InitiatingProcessSHA256)
12 | where parse_json(AdditionalFields).PipeName == @"\\Device\\Name"
13 | project Timestamp, DeviceId, DeviceName, ActionType, FileName
14 let Connection = DeviceNetworkEvents
15 | where ActionType == "ConnectionSuccess"
16 | where isnotempty(InitiatingProcessSHA256)
17 | join kind=inner (ImageLoads | distinct InitiatingProcessSHA256)
18 | project Timestamp, DeviceId, DeviceName, ActionType, RemoteIP
19 union NamedPipes, ImageLoads, Connection
20 | sort by Timestamp asc, DeviceId, InitiatingProcessSHA256
21 | scan with_match_id=Id declare (Step:string, Delta:timespan) w
22   step InitialConnection: ActionType == "ConnectionSuccess" =
23   step NamedPipe: ActionType == 'NamedPipeEvent' and DeviceId
24   step ImageLoad: ActionType == 'ImageLoaded' and DeviceId ==
25 )
26 | where Step == 's3'
27
```

Getting started Results Query history

Export Link to incident

Filters: Add filter

|                                     |                         |                       |
|-------------------------------------|-------------------------|-----------------------|
| <input checked="" type="checkbox"/> | Timestamp               | DeviceId              |
| <input checked="" type="checkbox"/> | > Oct 6, 2025 5:57:5... | 2694a7cc2225f3b66f... |
| <input checked="" type="checkbox"/> | > Oct 9, 2025 6:06:2... | 2694a7cc2225f3b66f... |
| <input checked="" type="checkbox"/> | > Oct 17, 2025 6:41:... | 2694a7cc2225f3b66f... |
| <input checked="" type="checkbox"/> | > Oct 25, 2025 2:18:... | 2694a7cc2225f3b66f... |

## Take actions

- Choose actions
- Choose target entities
- Review and submit

### Choose actions

Specify the actions you want to take. Only actions applicable to entities in the selected query results are available.

#### Devices

Isolation rules only apply to devices with specific client versions. See [documentation](#) for more information.

- ☐ Isolate device
- ☐ Collect investigation package
- ☐ Run antivirus scan
- ☐ Initiate investigation
- ☐ Restrict app execution

#### Files

- ☐ Quarantine file

#### Users

- ☐ Mark user as compromised
- ☐ Disable user
- ☐ Force password reset

#### Emails

- ☐ Move to mailbox folder
- ☐ Delete email
- ☐ Submit to Microsoft
- ☐ Initiate automated investigation

# File Actions

Required fields to map the files: SHA1, InitiatingProcessSHA1, SHA256 or InitiatingProcessSHA256  
DeviceId column is also required as device identifier.

The screenshot displays the Microsoft Defender XDR Advanced Hunting interface. The 'Take actions' panel is active, showing a list of actions to perform on the selected query results. The 'Files' section is expanded, showing the 'Quarantine file' action selected. The 'Choose file' dropdown is set to 'SHA1'. The 'Results' section shows a table of query results with columns for TimeGenerated and Timestamp.

| TimeGenerated         | Timestamp             |
|-----------------------|-----------------------|
| Oct 26, 2025 10:38... | Oct 26, 2025 10:38... |
| Oct 25, 2025 5:25...  | Oct 25, 2025 5:25...  |

# User Actions

Required fields to map the cloud account: AccountObjectId

- Mark user as compromised

Required fields to map the AD account: AccountSid

- Disable user
- Force password reset

**Advanced hunting**

New query\* × | New query\* × |

> Run query Last 30 days ▾

Query

```
1 IdentityLogonEvents
2 | where AccountName contain
3 | distinct AccountSid
```

Getting started Results Query history

Export ▾ Link to incident Take action

Filters: Add filter

AccountSid

S-1-5-21-4044...

**Take actions**

Choose actions

Choose target entities

Review and submit

Isolate device

Collect investigation package

Run antivirus scan

Initiate investigation

Restrict app execution

**Files**

Quarantine file

**Users**

Mark user as compromised

Disable user

Force password reset

# Email Actions

Required fields to map the files: NetworkMessageId, RecipientEmailAddress, EmailDirection and SenderFromAddress

Supported for EmailEvents and EmailPostDeliveryEvents table

The screenshot displays the 'Advanced hunting' interface in Microsoft Defender XDR. The left pane shows a KQL query for filtering email events based on domain and TypoSquatMax. The right pane, titled 'Take actions', contains a workflow diagram and a list of actions categorized by information, files, users, and emails. The 'Emails' category is highlighted with a dashed orange border and contains four actions: 'Move to mailbox folder', 'Delete email', 'Submit to Microsoft', and 'Initiate automated investigation'. A 'Next' button is visible at the bottom right of the 'Emails' section.

**Advanced hunting**

New query\* × | New query\* × |

Run query Last 30 days

**Query**

Query results are presented in your local timeline

```
1 let Domain = tolower("mic")
2 let UnicodeDomain = unicode(Domain)
3 let TypoSquatMin = 0.75;
4 let TypoSquatMax = 0.99;
5 EmailEvents
6 | where EmailDirection == "Out"
7 | extend SenderDomainUnicode = unicode(SenderFromAddress)
8 | extend TypoSquadPercent = (SenderDomainUnicode - UnicodeDomain) / (UnicodeDomain - Domain)
9 | where TypoSquadPercent >= TypoSquatMin and TypoSquadPercent <= TypoSquatMax
10 | project-reorder Timestamp, SenderFromAddress, RecipientEmailAddress, NetworkMessageId
```

Getting started **Results** Query history

Timeline visualization

Export Link to incident Telemetry

Filters: Add filter

|                                     | TimeGenerated           | Timestamp             |
|-------------------------------------|-------------------------|-----------------------|
| <input checked="" type="checkbox"/> | > Oct 14, 2025 4:00:... | Oct 14, 2025 4:00:... |
| <input checked="" type="checkbox"/> | > Oct 14, 2025 4:00:... | Oct 14, 2025 4:00:... |
| <input checked="" type="checkbox"/> | > Nov 2, 2025 7:27:3... | Nov 2, 2025 7:27:3... |
| <input checked="" type="checkbox"/> | > Nov 2, 2025 7:27:3... | Nov 2, 2025 7:27:3... |
| <input checked="" type="checkbox"/> | > Nov 2, 2025 7:27:3... | Nov 2, 2025 7:27:3... |
| <input checked="" type="checkbox"/> | > Nov 2, 2025 7:27:3... | Nov 2, 2025 7:27:3... |
| <input checked="" type="checkbox"/> | > Nov 2, 2025 2:47:4... | Nov 2, 2025 2:47:4... |

**Take actions**

Choose actions

Choose target entities

Review and submit

**Information**

- ☐ Isolate device
- ☐ Collect investigation package
- ☐ Run antivirus scan
- ☐ Initiate investigation
- ☐ Restrict app execution

**Files**

- ☐ Quarantine file

**Users**

- ☐ Disable user
- ☐ Force password reset

**Emails**

- ☐ Move to mailbox folder
- ☐ Delete email
- ☐ Submit to Microsoft
- ☐ Initiate automated investigation

Next

## Choose actions

Specify the actions you want to take. Only actions applicable to entities in the selected query results are available.

### Devices

ⓘ Isolation rules only apply to devices with specific client versions. See [documentation](#) for more information.

- ☐ Isolate device
- ☒ Collect investigation package
- ☐ Run antivirus scan
- ☒ Initiate investigation
- ☐ Restrict app execution

### Files

- ☒ Quarantine file

Choose file \*

InitiatingProcessSHA1

### Users

- ☐ Mark user as compromised
- ☒ Disable user

Choose user \*

AccountSid

- ☐ Force password reset

### Emails

- ☐ Move to mailbox folder
- ☐ Delete email
- ☐ Submit to Microsoft
- ☐ Initiate automated investigation

# Response Actions from Advanced Hunting

- Scalability
- Hunting
- IOC matching
- Perform multiple actions at once



# Be aware of your entity mapping!

- While building detections make sure the right columns are projected.
- Entity mapping is important for SOAR/AutoIR capabilities

**Custom detection**

General ☒ Alert settings ☐ Automated actions ☐ Scope ☐ Review and create ☐

**Entity**

Device

Identifier  HostName  Column  DeviceName

**Entity**

Device

Identifier  HostName  Column  RemoteDeviceName

**Entity**

User

Identifier  Sid  Column  AccountSid

**Entity**

User

Identifier  Sid  Column  InitiatingProcessAccountSid

**Entity**

Mailbox

Identifier  MailboxPrimaryAddress  Column  InitiatingProcessAccountUpn

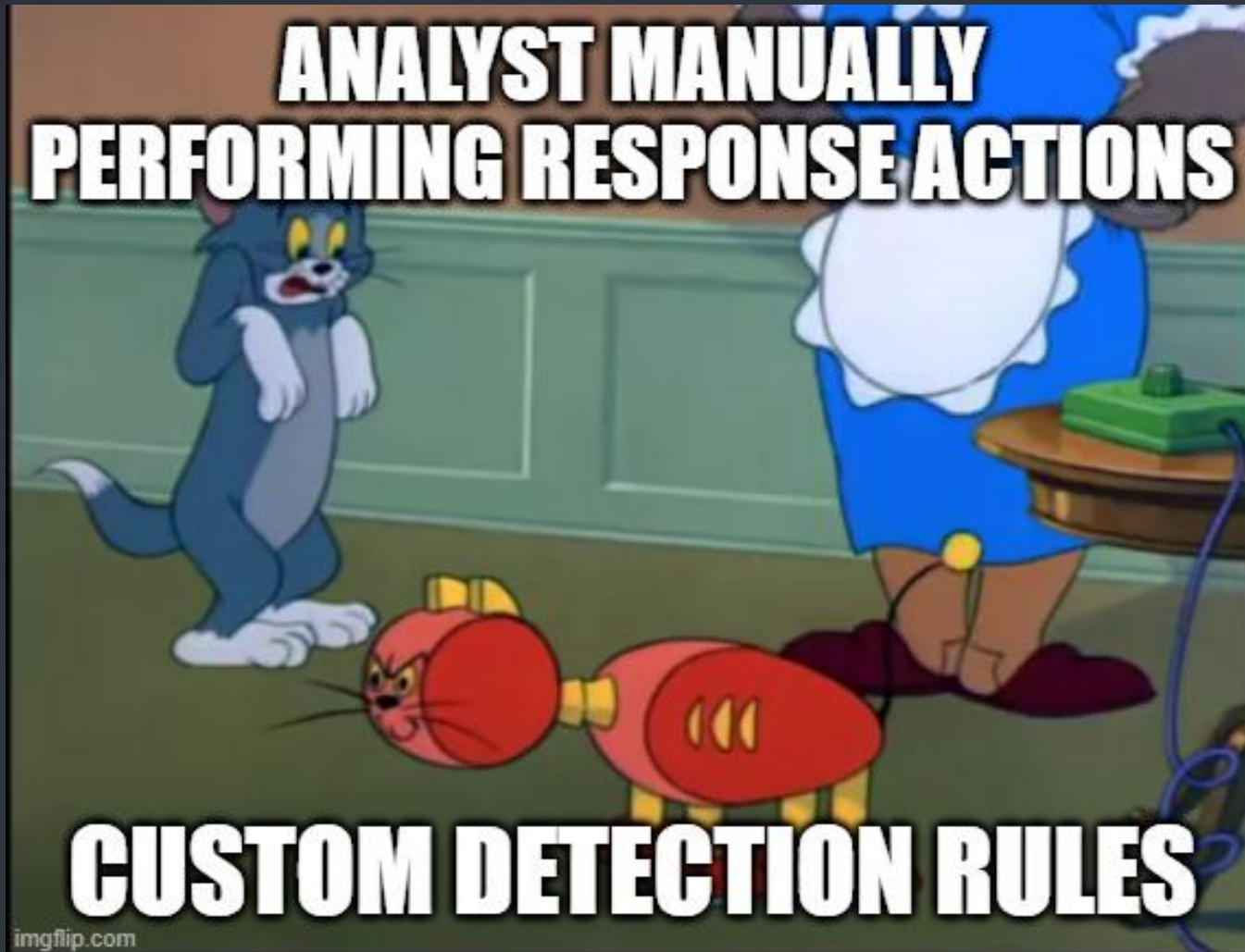
+ Add assets

Related Evidence

Entity

Back Next

**ANALYST MANUALLY  
PERFORMING RESPONSE ACTIONS**



**CUSTOM DETECTION RULES**

imgflip.com

# Custom Detections = AIR Tool

- Custom Detections have the capability to perform AutoIR
- Capabilities for Devices, Files, Users and Mails
- Use NRT when possible

# ID 99047: [DxBP][D1A0H0][MDE] Sliver C2 Beacon Detected

High | Active | Unassigned | Unclassified | Last update time: Nov 2, 2025 11:01 AM | ExcludedAttackDisruption

Go Hunt queries launched from the entity menu now default to a time range starting from the incident's start time up to the execution time. For incidents older than 30 days, the query defaults to last 30-day.

You can now manually ungroup entity groups in the graph, by clicking a group, open the menu, and select 'Ungroup'. To regroup, use the 'Group similar entities' toggle in the graph header.

Blast radius available for any entity. You can expand an entity's blast radius from the entity's menu options.

Attack story Alerts (1) Assets (2) Investigations (0) Evidence and Response (2) Summary Similar incidents (0)

## Detection & Categories

### Active alerts

1/1

### Categories

1

### First activity

Oct 6, 2025 5:57:58 PM

### Last activity

Oct 9, 2025 6:06:25 PM

### Creation time

Nov 2, 2025 11:01:48 AM

## Alerts



Oct 6, 2025 5:57 PM

New

[DxBP][D1A0H0][MDE] Sliver C2 Beacon Detected

workstation-03 Rick Astley

## Incident graph

Layout



Group similar nodes

## Advanced hunting

Run query [Share link](#)

Selected workspace: Sentinel [Set in query](#)

```
1 let ImageLoads = DeviceImageLoadEvents
2 | where ActionType == 'ImageLoaded'
3 | where FileName =~ "samlib.dll"
4 | where isnotempty(InitiatingProcessSHA256)
5 | invoke FileProfile(InitiatingProcessSHA256, 1000)
6 | where GlobalPrevalence <= 50 or isempty(GlobalPrevalence)
7 | project Timestamp, DeviceId, DeviceName, ActionType, FileName, InitiatingProcessFileName, InitiatingProcessSHA256, InitiatingProcessAccountSid, ReportId;
8 let NamedPipes = DeviceEvents
9 | where ActionType == 'NamedPipeEvent'
10 | where isnotempty(InitiatingProcessSHA256)
11 | join kind=inner (ImageLoads | distinct InitiatingProcessSHA256) on InitiatingProcessSHA256
12 | where parse_json(AdditionalFields).PipeName == @"\\Device\\NamedPipe\\wkssvc"
13 | project Timestamp, DeviceId, DeviceName, ActionType, FileName, InitiatingProcessFileName, InitiatingProcessSHA256, InitiatingProcessAccountSid, PipeName = parse_json(AdditionalFields).PipeName, ReportId;
14 let Connection = DeviceNetworkEvents
15 | where ActionType == "ConnectionSuccess"
16 | where isnotempty(InitiatingProcessSHA256)
17 | join kind=inner (ImageLoads | distinct InitiatingProcessSHA256) on InitiatingProcessSHA256
18 | project Timestamp, DeviceId, DeviceName, ActionType, RemoteIP, RemoteUrl, InitiatingProcessFileName, InitiatingProcessSHA256, InitiatingProcessAccountSid, ReportId;
19 union NamedPipes, ImageLoads, Connection
20 | sort by Timestamp asc, DeviceId, InitiatingProcessSHA256
21 | scan with_match_id=Id declare (Step:string, Delta:timespan) with (
22 |     step InitialConnection: ActionType == "ConnectionSuccess" => Step = "s1";
23 |     step NamedPipe: ActionType == 'NamedPipeEvent' and DeviceId == InitialConnection.DeviceId and InitiatingProcessSHA256 == InitialConnection.InitiatingProcessSHA256 and Timestamp between (Timestamp .. datetime_add('second', 1, InitialConnection.Timestamp)) and InitiatingProcessSHA256 == InitialConnection.InitiatingProcessSHA256 => Step = "s2";
24 |     step ImageLoad: ActionType == 'ImageLoaded' and DeviceId == NamedPipe.DeviceId and InitiatingProcessSHA256 == NamedPipe.InitiatingProcessSHA256 and Timestamp between (Timestamp .. datetime_add('second', 1, NamedPipe.Timestamp)) and InitiatingProcessSHA256 == NamedPipe.InitiatingProcessSHA256 => Step = "s3";
25 | )
26 | where Step == 's3'
```

Export [Show empty columns](#)

4 items

[Search](#)

00:05.62

Low

[Chart type](#)

[Full screen](#)

Filters: [Add filter](#)

|                          | Timestamp               | DeviceId              | DeviceName     | ActionType  | FileName   | InitiatingProcessFileName | InitiatingProcessSHA256 | InitiatingProcessAccountSid | PipeName | ReportId | RemoteIP | RemoteUrl |
|--------------------------|-------------------------|-----------------------|----------------|-------------|------------|---------------------------|-------------------------|-----------------------------|----------|----------|----------|-----------|
| <input type="checkbox"/> | > Oct 6, 2025 5:57:...  | 2694a7cc2225f3b66f... | workstation-03 | ImageLoaded | samlib.dll | teamviewer.exe            | 4cf757dfb93b919608...   | S-1-12-1-286326756...       |          | 6804     |          |           |
| <input type="checkbox"/> | > Oct 9, 2025 6:06:2... | 2694a7cc2225f3b66f... | workstation-03 | ImageLoaded | samlib.dll | teamviewer.exe            | 4cf757dfb93b919608...   | S-1-12-1-286326756...       |          | 16382    |          |           |
| <input type="checkbox"/> | > Oct 17, 2025 6:41:... | 2694a7cc2225f3b66f... | workstation-03 | ImageLoaded | samlib.dll | teamviewer.exe            | 4cf757dfb93b919608...   | S-1-12-1-286326756...       |          | 3737     |          |           |
| <input type="checkbox"/> | > Oct 25, 2025 2:18:... | 2694a7cc2225f3b66f... | workstation-03 | ImageLoaded | samlib.dll | jetbrains.exe             | e59f16b28d10b9b83...    | S-1-12-1-286326756...       |          | 44969    |          |           |

# Custom detection

- ✓ General
- ✓ Alert settings
- **Automated actions**
- Scope
- Review and create

## Automated actions

Define actions to automatically take on affected entities in the generated alerts and incidents.

### Remediation actions to take

Choose an applicable action to take on entities found by your query.

#### Devices



① Isolation rules only apply to devices with specific client versions. See [documentation](#) for more information.

- ☐ Isolate device
- ☐ Collect investigation package
- ☐ Run antivirus scan
- ☐ Initiate investigation
- ☐ Restrict app execution

#### Files



#### Users



#### Emails



# Custom detection

- ✓ General
- ✓ Alert settings
- **Automated actions**
- Scope
- Review and create

## Automated actions

Define actions to automatically take on affected entities in the generated alerts and incidents.

### Remediation actions to take

Choose an applicable action to take on entities found by your query.

#### Devices ▼

#### Files ^

- ☐ Allow/Block
- ☐ Quarantine file

#### Users ^

- ☐ Mark user as compromised
- ☐ Disable user
- ☐ Force password reset

#### Emails ^

- ☐ Move to mailbox folder
- ☐ Delete email

# Custom Detections = AIR Tool

## 98% TP rules

- Very low FP ratio
- High chance of compromise
  - High/Medium severity
  - Investigation Package

## Campaign Rules

- Campaign specific
- Temporary in nature
  - Written for AIR



# High Impact Alert Isolation

- Attack Disruption will act in some specific scenarios
- Depending on your risk appetite more isolation rules can be added
- NRT supported

```
1 AlertEvidence
2 | where EntityType in ('Machine', 'user')
3 | where Title has_any ('Ransomware', 'Hands-on-keyboard', 'RunMRU')
4 | where Severity == 'High'
5 | project Timestamp, Title, AlertId, Severity, DeviceId, DeviceName, AccountObjectId
6 |
```

Getting started **Results** Query history

Export Show empty columns

67 items Search

Filters: Add filter

| <input type="checkbox"/> | Timestamp               | Title                     | AlertId                 | Severity | DeviceId             | DeviceName      | AccountObjectId |
|--------------------------|-------------------------|---------------------------|-------------------------|----------|----------------------|-----------------|-----------------|
| <input type="checkbox"/> | > Oct 22, 2025 8:14:... | Suspicious command in ... | ⚡ da99c5be2d-f29f-44... | High     | 6d99a2b880a14561f... | kustocon-level1 | 👤               |
| <input type="checkbox"/> | > Oct 22, 2025 8:15:... | Compromised account c...  | ⚡ da08f3214e-19b2-43... | High     | 6d99a2b880a14561f... | kustocon-level1 | 👤               |
| <input type="checkbox"/> | > Oct 22, 2025 8:19:... | Compromised account c...  | ⚡ da08f3214e-19b2-43... | High     | 6d99a2b880a14561f... | kustocon-level1 | 👤               |
| <input type="checkbox"/> | > Oct 22, 2025 8:19:... | Compromised account c...  | ⚡ da08f3214e-19b2-43... | High     | 6d99a2b880a14561f... | kustocon-level1 | 👤               |

**It is not always  
clear**

**How can we help analysts  
in their decisions?**



# Incident triage/response

- Workbooks
- Logic/Function Apps
- KustoHawk

# Workbooks

- Standardize Incident Investigations
- Guides analysts
- Makes it easy to spot indicators
- Enrich alerts
- Visualizes data



# Workbook Demo

KUSTOCON

## Cloud Account Investigation

sentinel

Edit Open Save Refresh Add Remove ? Help Auto refresh: Off

Help me create my first Azure Workbook How do enterprises leverage Workbooks? Query recent Azure Monitor workbooks changes

Cloud Account Investigation

This workbook serves as tool to help simplify Cloud Account Investigations, such as AiTM, Anomalous Sign-Ins, Suspicious Tokens and post-compromise activities.

Subscription: All Workspace: All Timeframe: Last 90 days UserPrincipalName: rick.astley@kqlquery.com

Summary: rick.astley@kqlquery.com

| PrivilegedAccount | AccountUPN               | AccountDomain | Department | JobTitle            | Manager       | RiskLevel | RiskState | GroupMembership                                      | AssignedRoles |
|-------------------|--------------------------|---------------|------------|---------------------|---------------|-----------|-----------|------------------------------------------------------|---------------|
|                   | Rick.Astley@kqlquery.com |               | HR         | HR Business Partner | Bert-Jan Pals |           |           | ["All Internal Users","MSFT","Subscription Readers"] | []            |

Statistics

Alerts

| Severity      | Count |
|---------------|-------|
| Low           | 29    |
| Medium        | 22    |
| High          | 13    |
| Informational | 3     |

User Risk Events

| Count |
|-------|
| 4     |

Anomalies

The query returned no results.

# Logic Apps

- Incident Enrichment
- (Advanced) Response Actions
  - Wrapper around XDR Automated Response actions
  - Graph API Response actions
  - Third party Response actions
- IOC Integrations
- Function Apps



Details: <https://kqlquery.com/posts/logicapp-kql-api/Enrichment>:

- <https://kqlquery.com/posts/sentinel-automation-part1/>
- <https://kqlquery.com/posts/automatic-cisa-vulnerability-notifications/>
- <https://github.com/Bert-JanP/Sentinel-Automation>



# Kusto API Support

| Table Category                    | Azure Monitor API | Graph API | Graph API (Without Unified XDR) | Defender ATP API |
|-----------------------------------|-------------------|-----------|---------------------------------|------------------|
| Alerts & behaviors                | ✗                 | ✓         | ✓                               | ✗                |
| Apps & identities                 | ✗                 | ✓         | ✓                               | ✗                |
| Email & collaboration             | ✗                 | ✓         | ✓                               | ✗                |
| Devices                           | ✗                 | ✓         | ✓                               | ✓                |
| Defender Vulnerability Management | ✗                 | ✓         | ✓                               | ✓                |
| Email & collaboration             | ✗                 | ✓         | ✓                               | ✗                |
| Cloud Infrastructure              | ✗                 | ✓         | ✓                               | ✗                |
| Sentinel - Connector Data         | ✓                 | ✓         | ✗                               | ✗                |
| Sentinel - Custom Logs            | ✓                 | ✓         | ✗                               | ✗                |

# API Permissions

| API               | Application        | Permission             | Admin Consent |
|-------------------|--------------------|------------------------|---------------|
| Azure Monitor API | Log Analytics API  | Data.Read              | Required      |
| Graph API         | Graph              | ThreatHunting.Read.All | Required      |
| Defender ATP API  | WindowsDefenderATP | AdvancedQuery.Read.All | Required      |



# API Limitations

| Description                    | Azure Monitor API                                                          | Graph API                                                                                                            | Defender ATP API                                                         |
|--------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Timeframe                      | Table retention                                                            | Table retention                                                                                                      | 30 Days                                                                  |
| Max Rows                       | 500.000                                                                    | 100.000                                                                                                              | 100.000                                                                  |
| Calls                          | 200 requests per 30 seconds per Microsoft Entra user or client IP address. | You can make up to at least 45 calls per minute per tenant. The number of calls varies per tenant based on its size. | 45 calls per minute, and up to 1,500 calls per hour.                     |
| Query Execution Time (Seconds) | 600                                                                        | 180                                                                                                                  | 200                                                                      |
| Total Execution Time           | Not specified                                                              | 15-minute CPU cycles                                                                                                 | 10 minutes of running time every hour and 3 hours of running time a day. |

KustoHawk 🦅



- Triage & Response
- Defender XDR & Sentinel
- Device & Identity

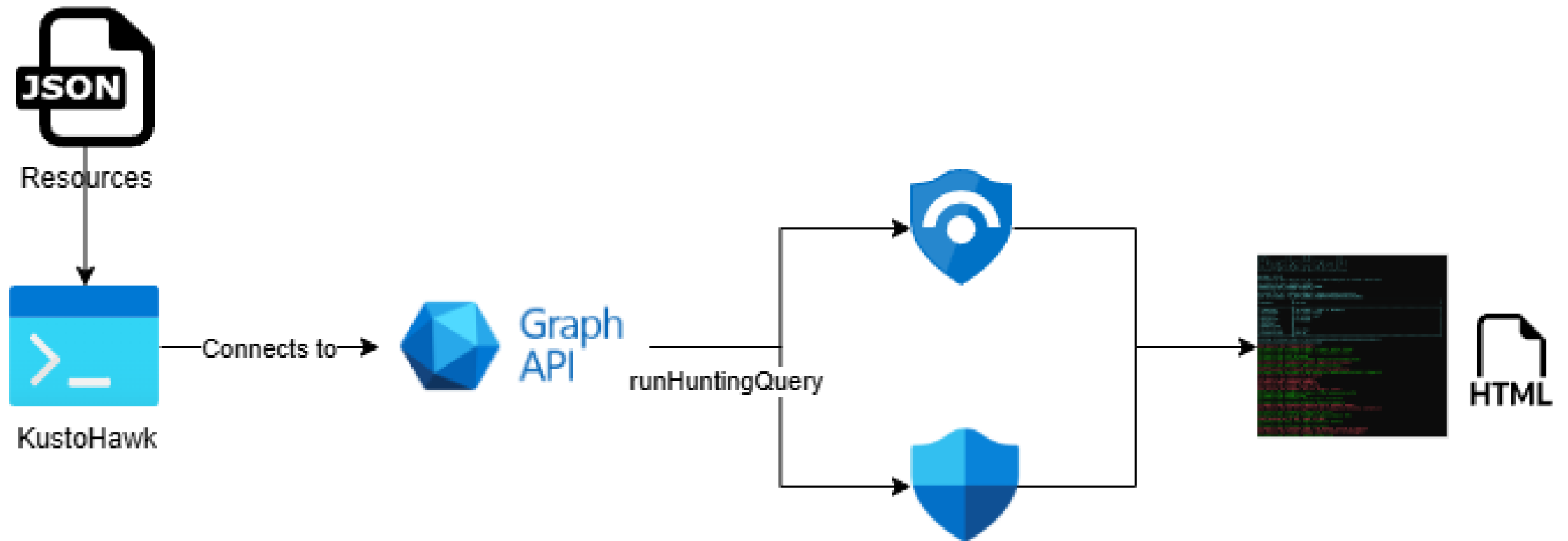
```
Version: 0.9.1
Developed by Bert-Jan Pals | Twitter: @BertJanCyber | Github: Bert-JanP
You selected User Authentication.
Connecting with Connect-GraphAPI-User...
Connecting to Microsoft Graph API...
DeviceId set to: 6d99a2b880a14561f58d9b4d1292b63cbc9a51ec
Host information for 6d99a2b880a14561f58d9b4d1292b63cbc9a51ec
```

| Details       | Values                       |
|---------------|------------------------------|
| Timestamp     | 2025-11-01T19:14:35.6835052Z |
| DeviceName    | kustocon-level1              |
| PublicIP      | 20.40.219.200                |
| OSPlatform    | Windows11                    |
| JoinType      |                              |
| AadDeviceId   |                              |
| MachineGroup  | KustoCon                     |
| ExposureLevel | Medium                       |

```
Starting 28 triage queries for 6d99a2b880a14561f58d9b4d1292b63cbc9a51ec
```

```
[39] Results for Triggered alerts
[0] Results for Executable files in users public folder
[0] Results for Executable files in ProgramData folder
[0] Results for AMSI triggers
[0] Results for Active CISA known exploited vulnerabilities
[3] Results for Suspicious RunMRU registry key entries
[0] Results for RMM tools with connections found
[2627] Results for Inbound connections from public ip
[0] Results for ASR events (excluding AsrLsassCredentialTheft triggers)
[12] Results for SmartScreen triggers
[25] Results for Antivirus events
[9] Results for Tampering attempts
[3] Results for Exploit Guard events
[12] Results for Registry Run and RunOnce events
[0] Results for Suspicious browser child processes events
[0] Results for MSHTA events
[0] Results for Anomalous SMB sessions identified
[0] Results for EDR configuration discovery events
[3] Results for Suspicious explorer child process events
[69] Results for New living of the land binary with external connection
[0] Results for Suspicious NamedPipe events
[0] Results for Abuse.ch ThreatFox malware domain hits
[346] Results for IP SUM (Level 4) hits
[0] Results for Rare .lnk file created on desktop
[0] Results for Defender exclusion events
[8] Results for Scheduled tasks from AppData created or updated
[39] Results for Process primary token elevated to SeDebugPriv
[0] Results for Potential beaconing activity
```

# KustoHawk



# KustoHawk 🦅 Demo



[illegible]

```
Host information for 2694a7cc2225f3b66f7cf8b6388a78b1857fadca
```

## Identity information for rick.astley@kglquery.com

| Details         | Values                      |
|-----------------|-----------------------------|
| Timestamp       | 2025-11-04T07:18:41.452769Z |
| AccountUpn      | Rick.Astley@kqlquery.com    |
| AccountName     | Rick Astley                 |
| RiskLevel       | None                        |
| Department      | HR                          |
| JobTitle        | HR Business Partner         |
| Manager         | Bert-Jan Pals               |
| PrivilegedRoles |                             |

Starting 30 triage queries for 2694a7cc2225f3b66f7cf8b6388a78b1857fadca

=====

[30] Results for Triggered alerts

[0] Results for Executable files in users public folder

[0] Results for Executable files in ProgramData folder

[0] Results for AMSI triggers

[3] Results for Active CISA known exploited vulnerabilities

[0] Results for Suspicious RunMRU registry key entries

[0] Results for RMM tools with connections found

[0] Results for Inbound connections from public ip

[0] Results for ASR events (excluding AsrLsassCredentialTheft triggers)

[6] Results for SmartScreen triggers

[3] Results for Antivirus events

[0] Results for Tampering attempts

[4] Results for Exploit Guard events

[0] Results for Registry Run and RunOnce events

[0] Results for Suspicious browser child processes events

[0] Results for MSHTA events

[0] Results for Anomalous SMB sessions identified

[1] Results for EDR configuration discovery events

[12] Results for Suspicious explorer child process events

[65] Results for New living of the land binary with external connection

[0] Results for Suspicious NamedPipe events

[0] Results for Rare .lnk file created on desktop

[5] Results for Defender exclusion events

[11] Results for Scheduled tasks from AppData created or updated

[7] Results for Process primary token elevated to SeDebugPriv

[1] Results for Potential beaconing activity

[3] Results for Sliver C2 beacon loaded

[0] Results for Abuse.ch ThreatFox malware domain hits

[0] Results for IP SUM (Level 4) hits

[0] Results for Twitter IOC hits (@0xDanielLopez)

Updated '.\Resources\DeviceQueries.json' with ResultCount values.

Report saved to C:\GitHub\KustoHawk\Reports\Device-ExecutedQueries-2694a7cc2225f3b66f7cf8b6388a78b1857fadca.html

Starting 17 traige queries for rick.astley@kqlquery.com

=====

[60] Results for Triggered alerts

[1] Results for User risk events

[0] Results for Potential AiTM sign-ins

[0] Results for Successful sign-in from suspicious user agent

[27] Results for Successful signin from new country

[68] Results for New user agent with signin attempts identified

[49] Results for New Entra ID application with signin attempts identified

[2] Results for Cloud persitence events

[2] Results for Successful device code sign-ins

[0] Results for Azure resources created

[0] Results for Azure resources deleted

[0] Results for Azure resources actioned

[0] Results for Graph \*.All permissions added

[0] Results for Graph ReadWrite permissions added

[0] Results for Conditional Access add, update, delete events

[0] Results for AzureHound activities

[0] Results for Email collection - Big Yellow Taxi 🚕

Updated '.\Resources\IdentityQueries.json' with ResultCount values.

Report saved to C:\GitHub\KustoHawk\Reports\Identity-ExecutedQueries-rick.astley@kqlquery.com.html



# KustoHawk

## GitHub:

- <https://github.com/Bert-JanP/KustoHawk>

# Conclusion

- Advanced Hunting helps to scale response actions
- Use custom detection rules to automate response actions
- Workbooks can standardize investigations
- Workbooks help analysts to focus on what matters
- KustoHawk 🦅 helps to triage and respond effectively

# Stickers



# Questions?

## KustoHawk 🦅:

- <https://github.com/Bert-JanP/KustoHawk>

## Cloud Investigation Workbook:

- <https://github.com/Bert-JanP/Sentinel-Automation/tree/main/Workbooks/Cloud%20Account%20Investigation>

## Sliver C2 Detection:

- <https://github.com/Bert-JanP/Hunting-Queries-Detection-Rules/blob/main/Defender%20For%20Endpoint/SliverC2BeaconLoaded.md>

## High Impact Alert Isolation:

- <https://github.com/Bert-JanP/Hunting-Queries-Detection-Rules/blob/main/Security%20Operations/NRT%20-%20AutoIRHighImpactAlert.md>



# Automating Incident Response with the Power of Kusto